Zixuan Jiang (Andrew)



Research Interests: Audio-Visual Multimodal Intelligence; Human–AI Interaction

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Education

Shanghai Innovation Institute & Shanghai Jiao Tong University 2027.09 (expected)

Ph.D. in Computer Science | School of Computer Science, Shanghai Jiao Tong University

Advisor: Xie Chen

Xi'an Jiaotong University 2023.09 – 2027.06 (expected)

Artificial Intelligence Experimental Class (AIH), Outstanding Talent Program | Qian Xuesen Honors College

Cumulative average: 92.26/100 (first five semesters); **94.78/100** (first six semesters).

Class rank: 5/66 (top 10%; unchanged across both periods).

Selected coursework: LLM 99, DSP 99, NLP 96, RL 96, CV 95. English: CET-6 575.

Young Gifted Program, Xi'an Jiaotong University 2021.09 – 2023.06

Qian Xuesen Honors College | **Class rank: 17/234 (top 8%)**

Research Experience

Shanghai Innovation Institute & Shanghai Jiao Tong University 2025.12 – Present

X-LANCE Lab | Advisor: Xie Chen, Associate Professor and Ph.D. Supervisor

- **Agentic speech recognition.** Co-first author of AgenticASR, which formulates speech recognition as final-intent-preserving audio-to-clean-text conversion and introduces a decoupled ASR–Refiner framework for bilingual online and offline refinement. The accompanying AASR-Bench contains **6,637 atomic rubric items**.
- **Interactive speech recognition.** Contributed to research on user feedback and semantic correction beyond one-pass ASR; the work introduces the S^2ER semantic-evaluation metric and an ISS simulation framework.
- **Audio generation and editing.** Contributed to benchmark design and construction and to the Agentic Prompt Enhancer for the Tencent Hunyuan AuK project; co-author of AuK and MMAE.

Xi'an Jiaotong University | School of Computer Science 2024.09 – Present

Advisor: Xiangyong Cao, Associate Professor and Ph.D. Supervisor

- **Remote-sensing multimodal interpretation.** Co-first author of DescribeEarth, published in **ISPRS P&RS**. The work introduces the Geo-DLC task, DE-Dataset, and DE-Bench for object-level fine-grained interpretation, together with scale-adaptive focal and domain-guided fusion modules.
- **Open-vocabulary perception.** Contributed to OVEarth-Bench, annotation-free remote-sensing segmentation, a survey accepted by the *Journal of Image and Graphics*, and the NTIRE 2026 cross-domain few-shot object detection challenge report.

Honors & Awards

- **National Scholarship** (2025, 2026); **Shuiyou First-Class Scholarship**, university-wide top 4 (2024); Outstanding Student, Xi'an Jiaotong University (2024, 2025); Third-Class Scholarship (2023); Qian Xuesen Honors College Awards for Innovation & Entrepreneurship and Academic Research/Competition.
- **Grand Prize**, Speech OPC Innovation & Entrepreneurship Competition (2026), awarded for VibeXASR, with Jun Guo, Tao Liu and Chonghao Cai. vibexasr.speech.wiki
- **ICPC Shaanxi Provincial Programming Contest**: Silver Medal (2025), Bronze Medal (2024). **National Lidar Conference Point Cloud Intelligent Analysis Competition**: Second Prize (2025).
- **National College Student Mathematical Modeling Contest, Shaanxi Division**: First Prize (2023, 2024, 2025). **Mathematical Contest in Modeling (MCM), USA**: Meritorious Winner (2024), Honorable Mention (2023).
- **China Mobile AI+ University Innovation Program, AI Track**: Third Prize (2024); the sole finalist team from Northwest China.

Projects & Leadership

MotionPilotX – Motion-Aware Intelligent Drone Control 2025.09 – 2026.06

Project Leader | Undergraduate Innovation and Entrepreneurship Project | Advisor: Longjun Liu

Led the development of a YOLOv11-based object detection and motion-pose regression system on Orange Pi RK3588 for on-device UAV perception and control. Bronze Prize, Entrepreneurship Track, Tengfei Cup Innovation and Entrepreneurship Competition, Xi'an Jiaotong University.

Leadership. Head of the Scientist Spirit Drama Troupe, Qian Xuesen Honors College Student Union (2025); member of the Qian Xuesen Honors College AI Study Group; presented and participated in discussions on large language models at the DeepSeek Youth Exchange Event.

Selected Publications

* Equal contribution. † Corresponding author. Author lists are abbreviated where necessary; complete author lists are available in the linked papers.

Remote Sensing Image Interpretation

DescribeEarth: Detailed localized captioning for remote sensing images (ISPRS P&RS, 2026, Vol. 242, pp. 860–872)
Kaiyu Li*, **Zixuan Jiang***, Xiangyong Cao†, ..., Zhi Wang

The Second Challenge on Cross-Domain Few-Shot Object Detection at NTIRE 2026: Methods and Results (CVPRW 2026)
Xingyu Qiu, Yuqian Fu, Jiawei Geng, ..., **Zixuan Jiang**, ...

Advances in Open-Vocabulary Perception for Remote-Sensing Images (Journal of Image and Graphics, 2026, accepted)
Kaiyu Li, Xiangyong Cao†, **Zixuan Jiang**, Deyu Meng

Agentic Speech Recognition

AgenticASR: Refining Speech Recognition in Real-World Scenarios via an Agentic Approach (AAAI 2027, under review)
Zixuan Jiang*, Binghao Qiang*, Jiaying Chi*, Yanqiao Zhu, Kai Yu, Xie Chen†

Towards Human-Like Interactive Speech Recognition With Agentic Correction and Semantic Evaluation (Speech Communication, under review)
Zixuan Jiang*, Yanqiao Zhu*, Peng Wang*, ..., Xie Chen†

Audio Generation & Editing

AuK Technical Report: An Open-Source Foundational Model for Speech Generation and Editing (Technical Report)
Ziyang Ma, Zhikang Niu, Wenming Tu, ..., **Zixuan Jiang**, ..., Xie Chen

MMAE: A Massive Multitask Audio Editing Benchmark (ICLR 2027, under review)
Ziyang Ma, Ruiqi Yan, Ruiyang Xu, ..., **Zixuan Jiang**, ..., Xie Chen